

Problem 25.12

Find an expression for t , $0 < t < 1$, such that 1 paid at time t is equivalent to 1 paid continuously between 0 and 1.

Solution

The present value of a payment of 1 made at time t is

$$v^t.$$

The present value of 1 paid continuously between 0 and 1 is the present value of a continuous annuity of rate 1 for one year:

$$\bar{a}_{\overline{1}|} = \frac{1 - v}{\delta}.$$

For the two payments to be equivalent, their present values must be equal:

$$v^t = \frac{1 - v}{\delta}.$$

Taking natural logarithms of both sides gives

$$\ln\left(\frac{1 - v}{\delta}\right) = \ln(v^t).$$

Since

$$\ln(v^t) = t \ln(v),$$

we have

$$\ln\left(\frac{1 - v}{\delta}\right) = t \ln(v).$$

Solving for t ,

$$t = \frac{\ln\left(\frac{1-v}{\delta}\right)}{\ln(v)}.$$

Thus,

$$\boxed{t = \frac{\ln\left(\frac{1-v}{\delta}\right)}{\ln(v)}}.$$